

1. Describe your research scenario and question(s)

The research scenario I am exploring with my dataset is that if there is any correlation between Gre Scores and Chance of Admit in a College for a student.

The Graduate Record Examination (GRE) is a standardized test used as an admission requirement for many graduate programs in the United States and other countries. The test assesses verbal reasoning, quantitative reasoning, and analytical writing skills. GRE scores are often used by universities as a predictor of academic success in graduate school.

To understand the relationship between GRE scores and chance of admission, several studies have been conducted. One such study found that there is a positive correlation between GRE scores and graduate school admission. The study showed that students with higher GRE scores had a higher chance of admission, but the correlation was not perfect, as there were cases where students with lower GRE scores were admitted over those with higher scores.

A common question for many prospective graduate students is what GRE score is needed for admission to a particular program. While there is no one-size-fits-all answer, it is generally accepted that a higher GRE score increases one's chances of admission. However, there are other factors such as academic performance, letters of recommendation, and personal statements that also play a role in the admissions process.

That is what I will be Testing.

2. Describe the data set

The Graduate Record Examination (GRE) is a standardized test used as an admission requirement for many graduate programs in the United States and other countries.

GRE scores are often used by universities as a predictor of academic success in graduate school.

The chance of admit refers to the probability or likelihood of being accepted into a college or university.

The Test of English as a Foreign Language (TOEFL) measures the English language proficiency of non-native English speakers.

University rating is a system of evaluating and ranking universities based on various factors such as academic performance, research output, faculty quality, and student satisfaction.

A statement of purpose (SOP) is a document that outlines an applicant's academic and professional background, goals, and reasons for applying to a particular program.

A letter of recommendation (LOR) is a document written by someone who can vouch for an applicant's skills, abilities, and character, typically used in the application process for jobs, internships, or higher education.

CGPA (Cumulative Grade Point Average) is a measure of academic performance that calculates the average grade point of a student across all courses taken over a certain period.

Research is whether the student has done any research papers or not, this can benefit in favor of acceptance from the college that the student chooses to apply.

Classification: - The chance of admit will be based on the GRE, TOEFL, SOP, LOR, university rating, CGPA

Number of tuples: - 9

Name of the attribute	Attribute description
GRE	Marks student obtained in the GRE exam
TOEFL	Marks student obtained in the TOEFL exam
UNIVERSITY RATING	Rating of the university that the student has applied to
SOP	Statement Of Purpose Letter provided by the student
LOR	Letter of recommendation provided by the student
CGPA	Marks student obtained in the high school examination
RESEARCH	Research Paper that the student has worked on
CHANCE OF ADMIT	Chances that the student will be accepted in the university

Link: - <https://github.com/kiat/R-Examples/tree/master/projects/datasets>

R-Examples/projects/datasets/project-83.csv

The data set is attached with the zip file along with the project proposal.

(Name of attached Data Set: - MachineLearningProjectDataSet)

3. Describe the statistical methods you plan to use

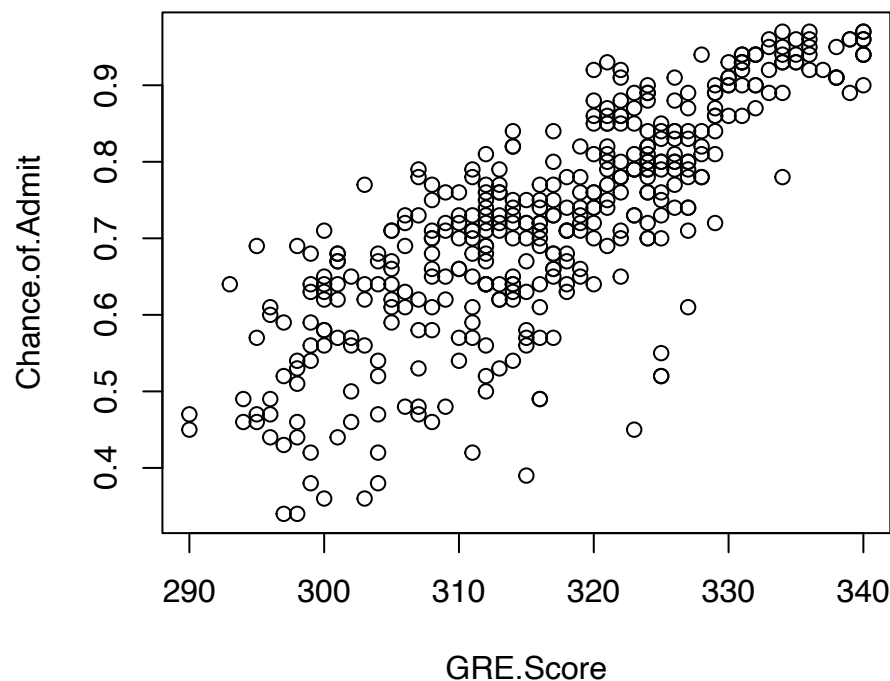
F-test method for simple linear regression between GRE scores and Chance of Admit.

4. Report your results

#Data-SET Exploration

The first test is a Sample Scatter Plot.

What is wanting to see what the plot of Gre Scores VS. Chance of Admit looks like.



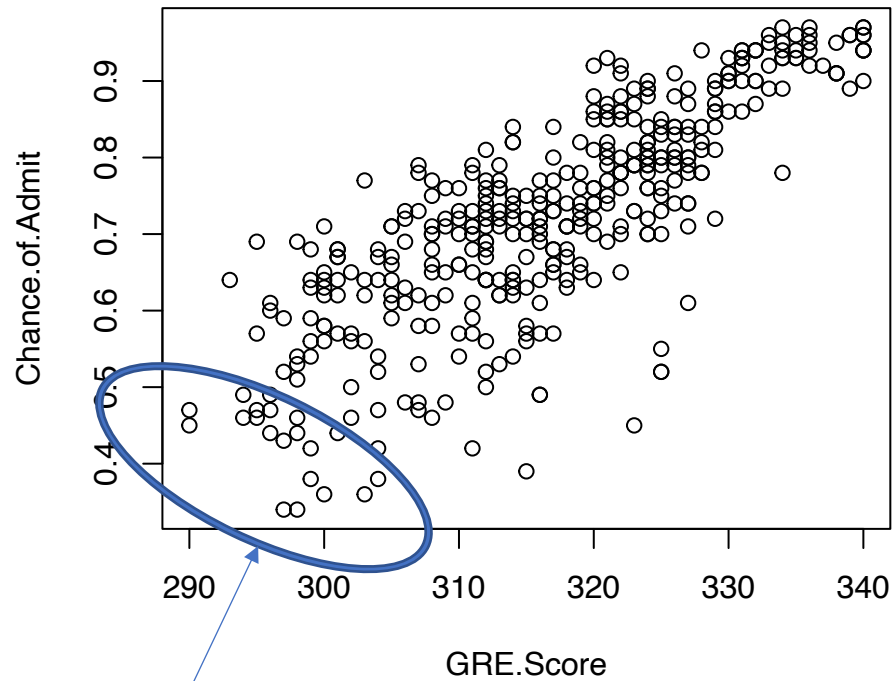
Next, let's look at the correlation coefficient to see how correlated these two variables are. Correlation: 0.8026105

Therefore about 80% Correlation

From the Scatter Plot, this looks a bit interesting.

There is may be a Positive Correlation?

Also, I suspect a few Outliers.

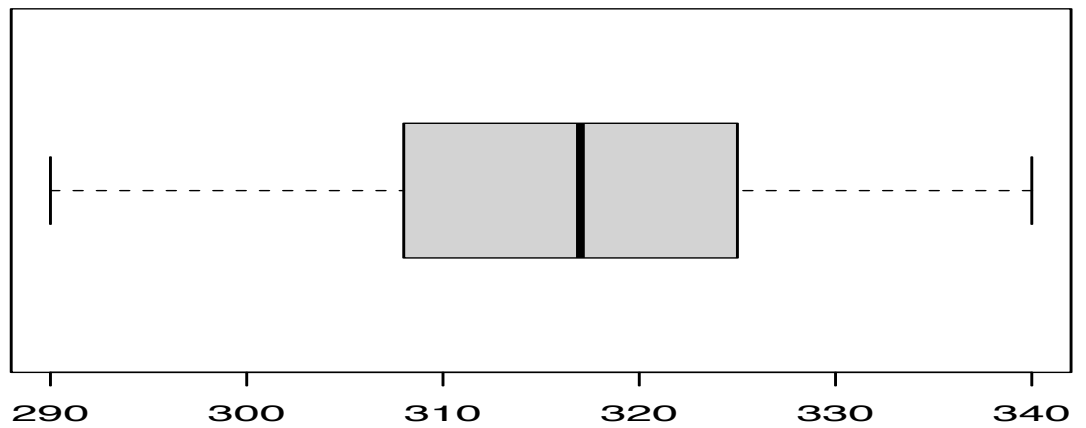


Maybe the outliers are here?

Let's try the Box Plot to see if there any outliers.

Five Number Summary of GRE Score: -

Minimum	Q1	Median	Q3	Maximum
290	308	317	325	340

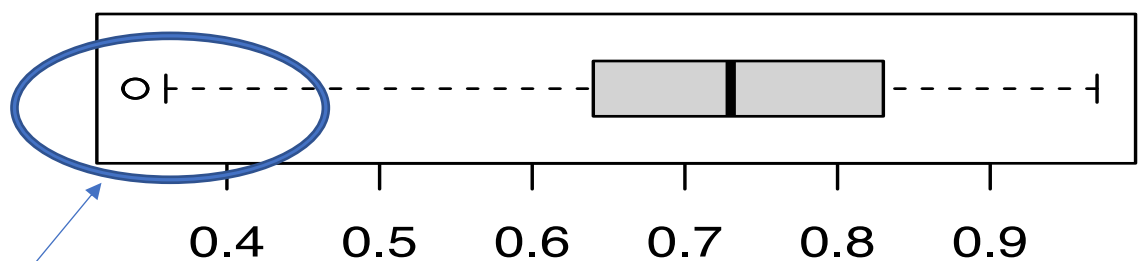


Well, Nothing Here.

How about Chance of Admit?

Five Number Summary of Chance of Admit: -

Minimum	Q1	Median	Q3	Maximum
0.34	0.64	0.73	0.83	0.97



Well, we do have an Outlier here.

Assumption: -

There are several possible assumptions as to why a college or university may accept a student with lower GRE scores. Here are a few:

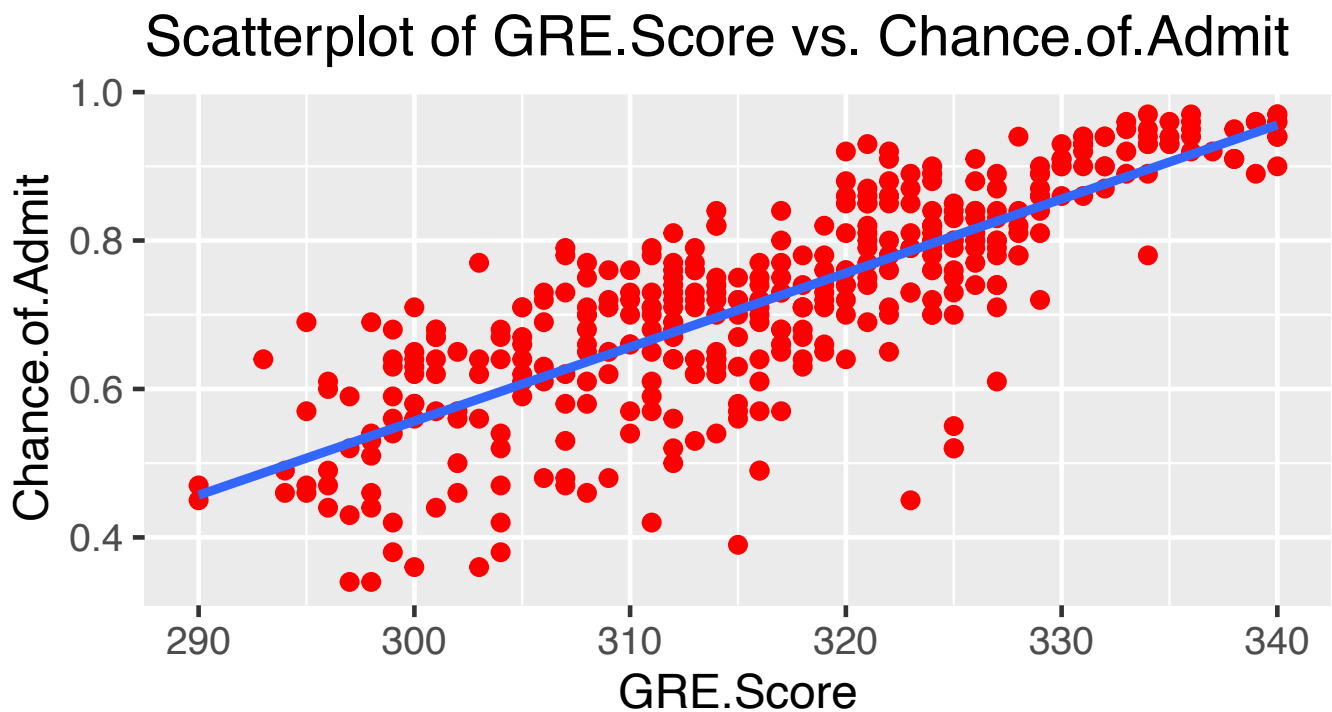
1. Holistic admissions: Many colleges and universities take a holistic approach to admissions, which means that they consider a variety of factors beyond just test scores. They may also look at a student's GPA, letters of recommendation, personal statement, extracurricular activities, and other factors when making admissions decisions. In this case, a student with lower GRE scores may still be accepted if they excel in other areas.
2. Test-optional policies: Some colleges and universities have adopted test-optional policies, which means that they do not require GRE scores as part of the admissions process. In this case, a student with lower GRE scores may choose not to submit their scores at all, and may still be accepted based on other factors.
3. Program-specific requirements: Depending on the program of study, some colleges and universities may place less emphasis on GRE scores than others. For example, a program in the humanities may be more interested in a student's writing ability and critical thinking skills, while a program in the sciences may be more interested in a student's quantitative abilities. In this case, a student with lower GRE scores may still be accepted if their strengths align with the program's requirements.
4. Limited pool of applicants: In some cases, a college or university may not receive enough applications from highly qualified students, so they may need to lower their admissions standards in order to fill their class. In this case, a student with lower GRE scores may still be accepted if they are among the most qualified applicants in the pool.

F-Test: -

In this test is where I will be tackling F-test method for simple linear regression between GRE scores and Chance of Admit.

The before sample tests were to provide a background to the F-Test Method.

The F-test is a statistical method used to compare the variances of two or more populations, and the before sample tests were conducted to establish a baseline or background data for the populations being compared using the F-test method.



Now, we can find the least squares regression equation which is equal to the following:

$$Y = 0.009975882 * (\text{GRE Scores}) - 2.436084$$

A Strong Positive Correlation Plot

Correlation: 0.8026105

Therefore about 80% Correlation

ANOVA TABLE

	DF	SUM SQ	MEAN SQ	F-VALUE	PR(>F)
GRE SCORE	1	5.2273	5.2273	720.55	< 2.2e-16
RESIDUALS	398	2.8873	0.0073		

RESIDUALS

Minimum	Q1	Median	Q3	Maximum
-0.33613	-0.04604	0.00408	0.05644	0.18339

COEFFICIENTS

	Estimate	Std. Error	t value	PR(> t)
(Intercept)	-2.4360842	0.1178141	-20.68	<2e-16
GRE.Score	0.0099759	0.0003716	26.84	<2e-16

Residual standard error: 0.08517 on 398 degrees of freedom

Multiple R-squared: 0.6442, Adjusted R-squared: 0.6433

F-statistic: 720.6 on 1 and 398 DF, p-value: $< 2.2e-16$

Step 1: Setup hypotheses and select alpha level

$B_1 = 0$ (there is no linear association) & $B_1 \neq 0$ (there is a linear association)

Alpha level = 0.05

`confint(m, level=0.95)`

	2.5 %	97.5 %
(Intercept)	2.667700007	-2.2044685
GRE.Score	0.009245267	0.0107065

Step 2: Select the appropriate test statistic.

$F = \text{RegMS} / \text{ResMS with 1 and } n-2 = 398 \text{ degrees of freedom}$

Step 3: State the decision rule

$qf(.95, df1=1, df2=398) = 3.864929$

Step 4: State the test statistic

$F_Statistic <- 5.2273 / 0.0073 = 720.6$

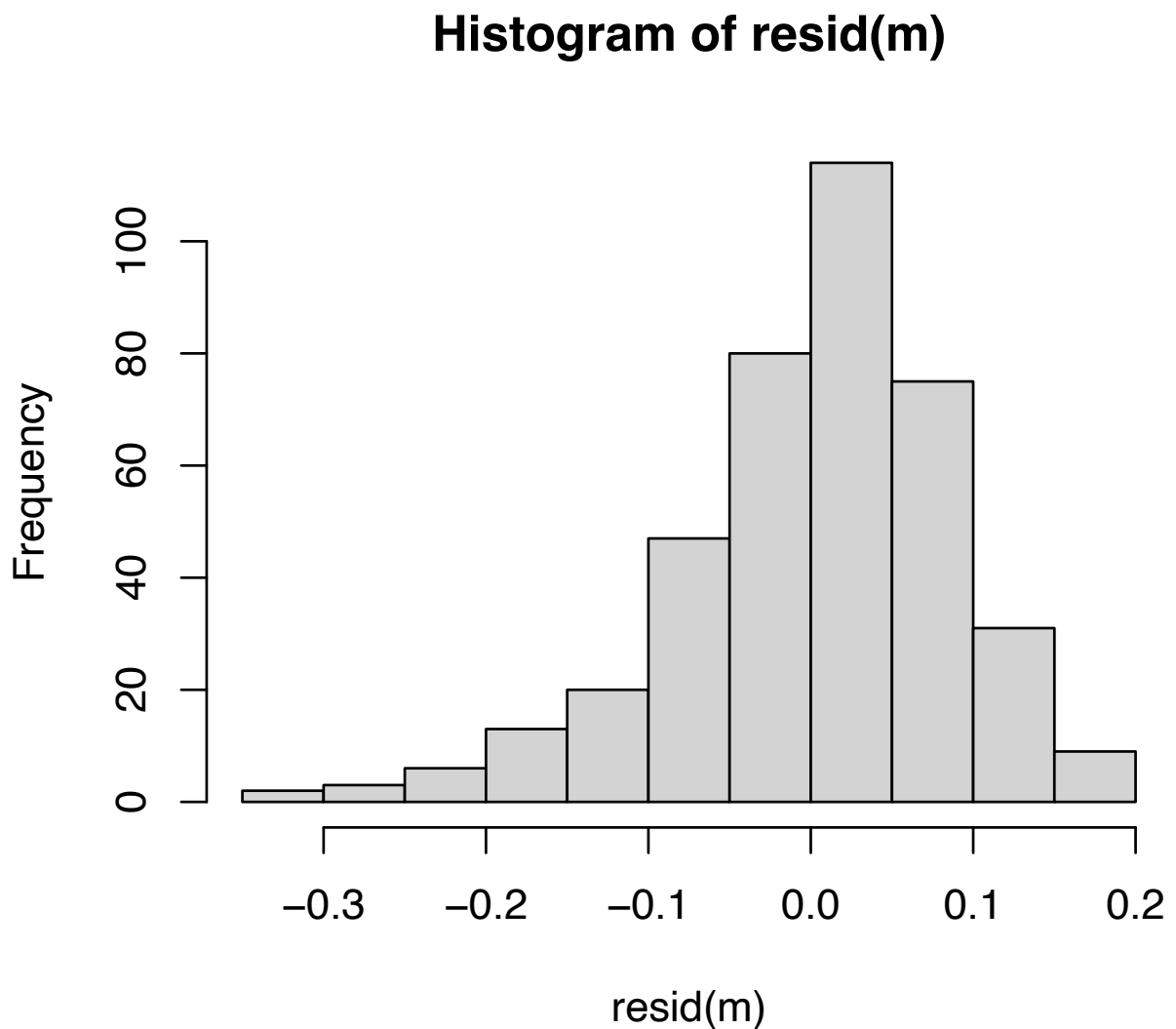
p-value: $2.2e-16$

Step 5: Conclusion

Reject Null Hypothesis.

We have significant evidence at the $\alpha=0.05$ level that $\beta_1 \neq 0$.

There is evidence of a significant linear association between GRE Scores and Chance of Admit.



Looks like the Histogram is Left Skewed.

Residual Plots and Using the Weight Option `lm()` to improve the model Further.

```
> lm(m)
```

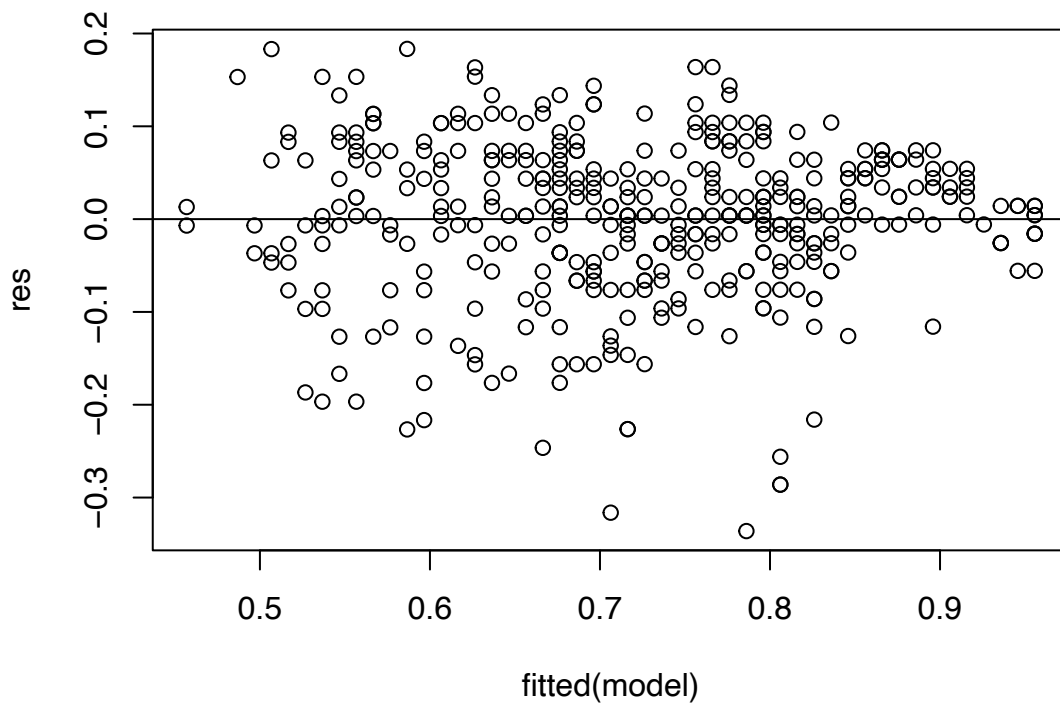
```
Call:
```

```
lm(formula = m)
```

```
Coefficients:
```

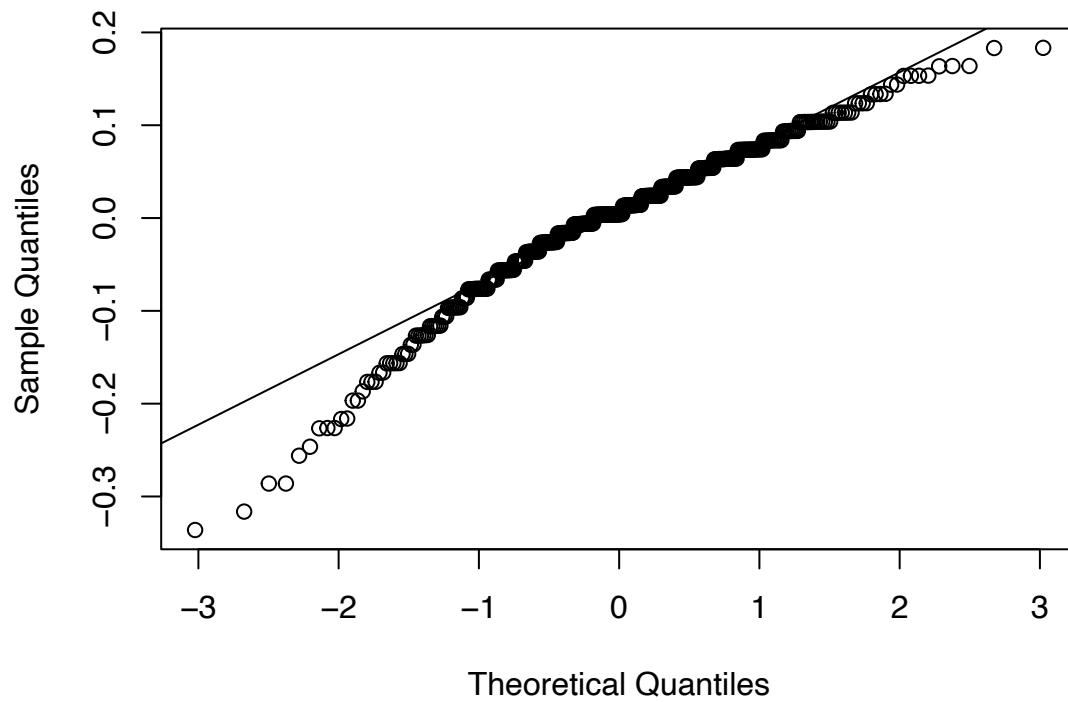
(Intercept)	student\$GRE.Score
-2.436084	0.009976

Produce residual vs. fitted plot with a horizontal line at 0.



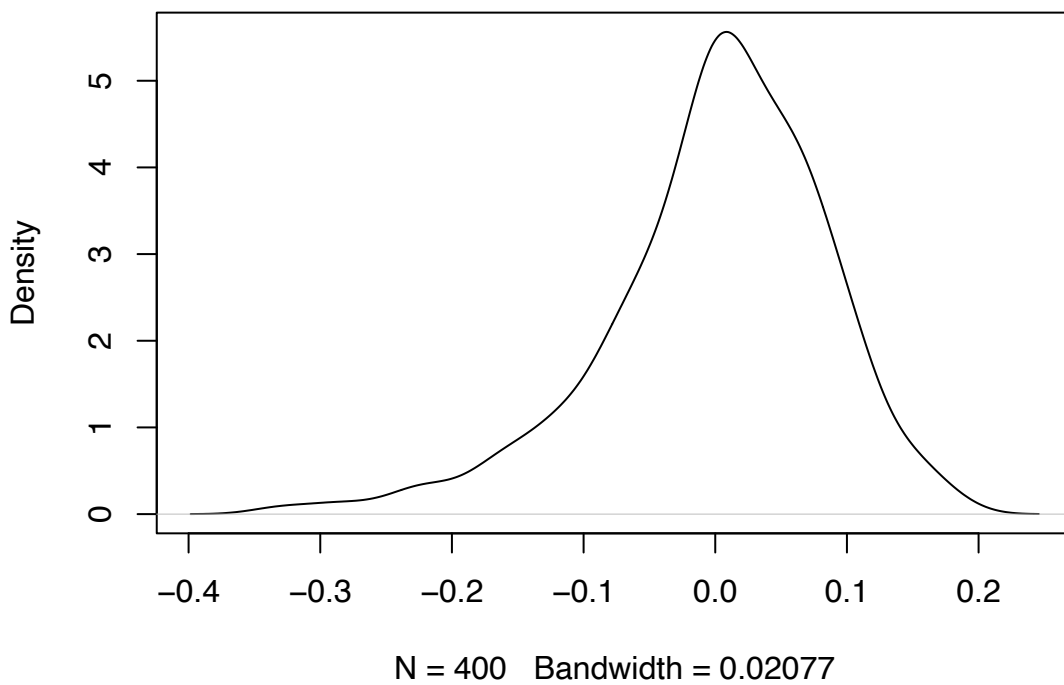
Creating a Q-Q plot for residuals with a Straight diagonal line to the plot

Normal Q-Q Plot



Producing a density plot

density.default(x = res)



Model Assumptions:

Linearity: As we saw from the scatter plot above, the model does meet the linearity assumption.

Independence: As we know in our data, we are collecting data on individuals only once, so our assumptions are met here that there is independence.

Normality: We can see that the residuals follow a normal distribution (Left Skewed), so our assumptions are met here.

Variance: We know that the equal variance assumption is also met because our model residual values aren't widely spread out in a triangular fashion.

5. State your conclusions and discuss any limitations

- As we can see from the F-Tests that we performed above that we have evidence against the null hypotheses in our tests.
- F-test we wanted to analyze to see for simple linear regression.
- Now that I did some baseline level analysis to give us a rough idea of the correlation between GRE Scores and Chance of Admit.
- My model indicated that there was significant evidence as there is a Positive Linear Association which tells us that as Students with better GRE Scores have a better chance of Admit.
- Lastly, I ran a linear regression histogram. After running my analysis, I concluded that this leaves me to believe that my data is reliable in helping us understand the process of college acceptance.
- I feel the analysis does have some limitations in that the sample size is not very large, but its large enough that we can make some assessments.
- I could get more data sets and maybe even broaden the applicant pool to younger adults and see how that would affect our overall results.
- Another study found that GRE scores were not the most important factor in predicting graduate school success. The study showed that undergraduate GPA and letters of recommendation were better predictors of academic success in graduate school than GRE scores.
- It is important to note that while a high GRE score may increase one's chances of admission, it is not the only factor considered by admissions committees. The overall profile of the applicant, including academic

performance, work experience, and other factors, will also be taken into consideration.

- In conclusion, GRE scores do play a role in the graduate school admissions process, but they are not the only factor considered. While a higher GRE score may increase one's chances of admission, it is important to have a well-rounded application that showcases academic achievement, work experience, and other relevant factors. Thank you for listening to my presentation.